

Raghavendra Mishra

Email : rm8166660@gmail.com

LinkedIn : [linkedin.com/in/raghavendra-mishra-iit-bbs](https://www.linkedin.com/in/raghavendra-mishra-iit-bbs)

Phone: (+91) 7054808227

IIT Bhubaneswar

Mechanical System Design

M.Tech Final Year

CGPA: 8.20/10



Education

Year	Degree/Examination	University/Board	%/CGPA
2026	M.Tech in Mechanical System Design	Indian Institute of Technology Bhubaneswar	8.20/10
2023	B.Tech in Mechanical Engineering	Rajkiya Engineering College Mainpuri	7.92/10
2018	Intermediate (12 th standard)	Uttar Pradesh Board	80.8%
2016	Secondary School (10 th standard)	Uttar Pradesh Board	87.5%

Internship Experience

Banaras Locomotive Workshop Varansi

1 Jul– 28 Jul 2022

Mechanical Engineering Intern

- **TMS** Managed workflow from incoming materials to component machining, assembly, and final checks, gaining understanding of manufacturing accuracy, tolerances, and quality requirements.
- **SAS** Worked on machining, cleaning, and assembling key engine and locomotive components while interpreting drawings, ensuring correct fit up, and supporting quality inspections.
- **LTS** Learned comprehensive electrical, engine, brake, load, and predelivery testing procedures, including performance data interpretation, deviation understanding, and final quality validation.

Projects

Micromechanical model for predicting Effective Thermo Electro Mechanical properties of Piezoelectric Composites

Master's Thesis

- Developed an analytical micromechanics based model to accurately predict the coupled Thermo Electro Mechanical (TEM) behavior of 1-3 and 1-3-2 piezoelectric fiber reinforced composites under thermal and mechanical loading.
- Developing the Micromechanics Modal for Predicting Thermo Electro Mechanical property when Both Fibre and Matrix Piezoelectrically Active.
- Investigating piezoelectric materials with High Temperature capability, with planned experimental validation of their thermo electro mechanical stability.

Composite development of Al6061 with Mg, Zn, and marble dust using Friction Stir Processing.

Bachelor's Thesis

- Enhance tensile, compressive, hardness, and microstructural properties of Al6061 composites.
- Develop lightweight, high performance materials via Friction Stir Processing.

Course Projects

Numerical Analysis of Single Lap Pretension Bolted Joint.

Jan 2025– May 2025

- Analyzed load capacity of single lap bolted joints under varying pretension and friction conditions.
- Conducted comprehensive static structural analysis using ANSYS Workbench to evaluate mechanical performance.
- Generated comparative plots and tables to present maximum and minimum value variations in deformation and stress.

Stress and Vibration Analysis Of The Wind Turbine Blade.

Jul 2024– Dec 2024

- Conducted a comprehensive study of the stresses and deformations of a wind turbine blade under steady-state conditions.
- Compared the performance of three distinct materials: E-glass fiber, Kevlar, and Carbon Fiber Reinforced Plastic (CFRP).
- Analyzed and visualized deformation patterns (mode shapes) across different vibration frequencies.

Dynamic Vibration Absorber.

Jul 2024– Dec 2024

- Demonstrate that the primary mass displacement is minimized when the absorber's natural frequency matches the excitation frequency.
- Observed how the absorber moves 180° out of phase to cancel primary system forces.
- Generate a frequency response function (FRF) to visualize the "anti-resonance" valley between the two new peaks.

Key Courses Taken _____

Vibrations, Finite Element Method in Engineering, Modal Analysis, Advanced Solid Mechanics

Technical Skills _____

CAD & Design: SolidWorks (Part, Assembly, Drawing Sheet GDT), AutoCAD, Fusion 360

Simulation Tools: ANSYS Workbench (Static, Modal, Harmonic), ANSYS APDL, Abaqus

Programming: Matlab, Python

Positions of Responsibility _____

Institute of Technology, Bhubaneswar

Jul 2025– Present

Teaching Assistantship (TA)

- Finite Element Method in Engineering (Theory Course)
- Machine Design Practice (Practical Course)
- Material Testing Lab
- Engineering Mechanics (Theory Course)

Hobbies _____

Playing Cricket, Playing Video Games

Languages _____

English: Professional Working Proficiency

Hindi: Native / Bilingual Proficiency